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OT-2 Simulation Protocol Report

Summary

Pipetting steps:	**192**
Labware items:	**3**
Tracked plates:	**1**
Deck snapshots:	**193**

Labware Layout

Slot 1: Opentrons OT-2 96 Tip Rack 30…

Slot 2: NEST 96 Well Plate 200 µL Flat

Slot 4: NEST 1 Well Reservoir 195 mL

Deck Layout

Slot 10 –	Slot 11 –	Trash –
Slot 7 –	Slot 8 –	Slot 9 –
Slot 4 NEST 1 Well Reservoir 1…	Slot 5 –	Slot 6 –
Slot 1 Opentrons OT-2 96 Tip R…	Slot 2 NEST 96 Well Plate 200 …	Slot 3 –

Generated by OT-2 Simulation Log Visualizer • BIOE234 Final Project

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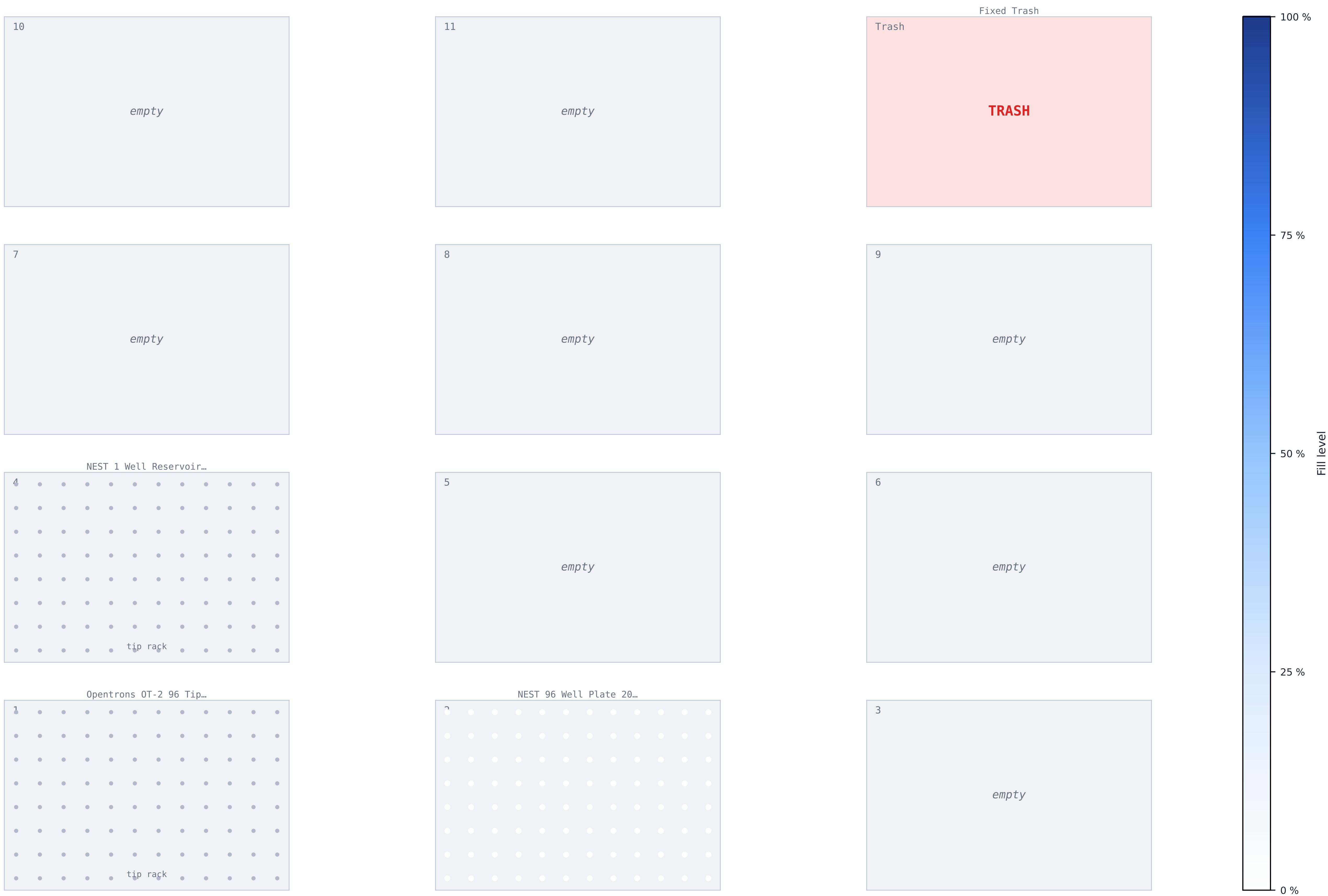
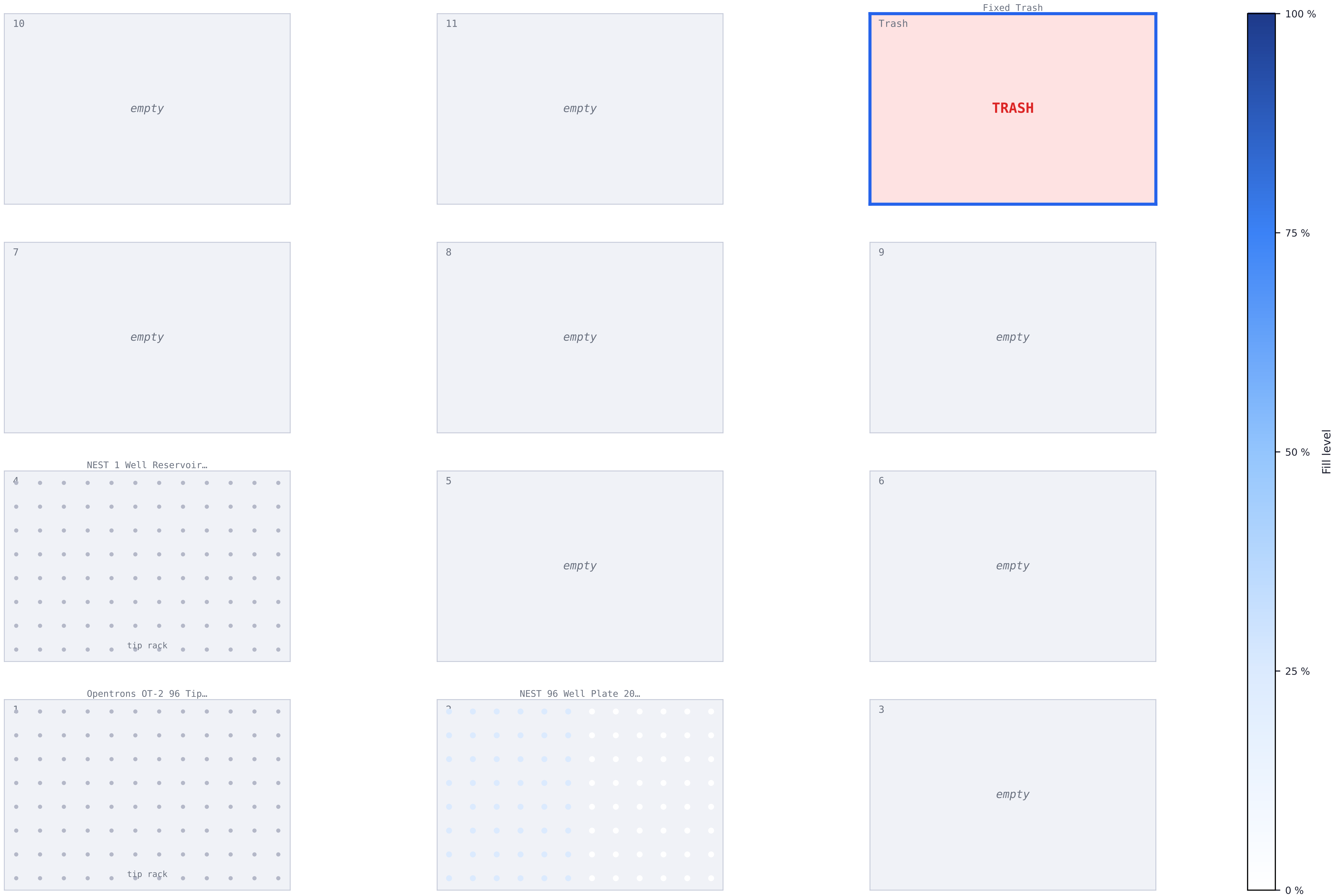


test · Step 191: Drop tip at slot 12



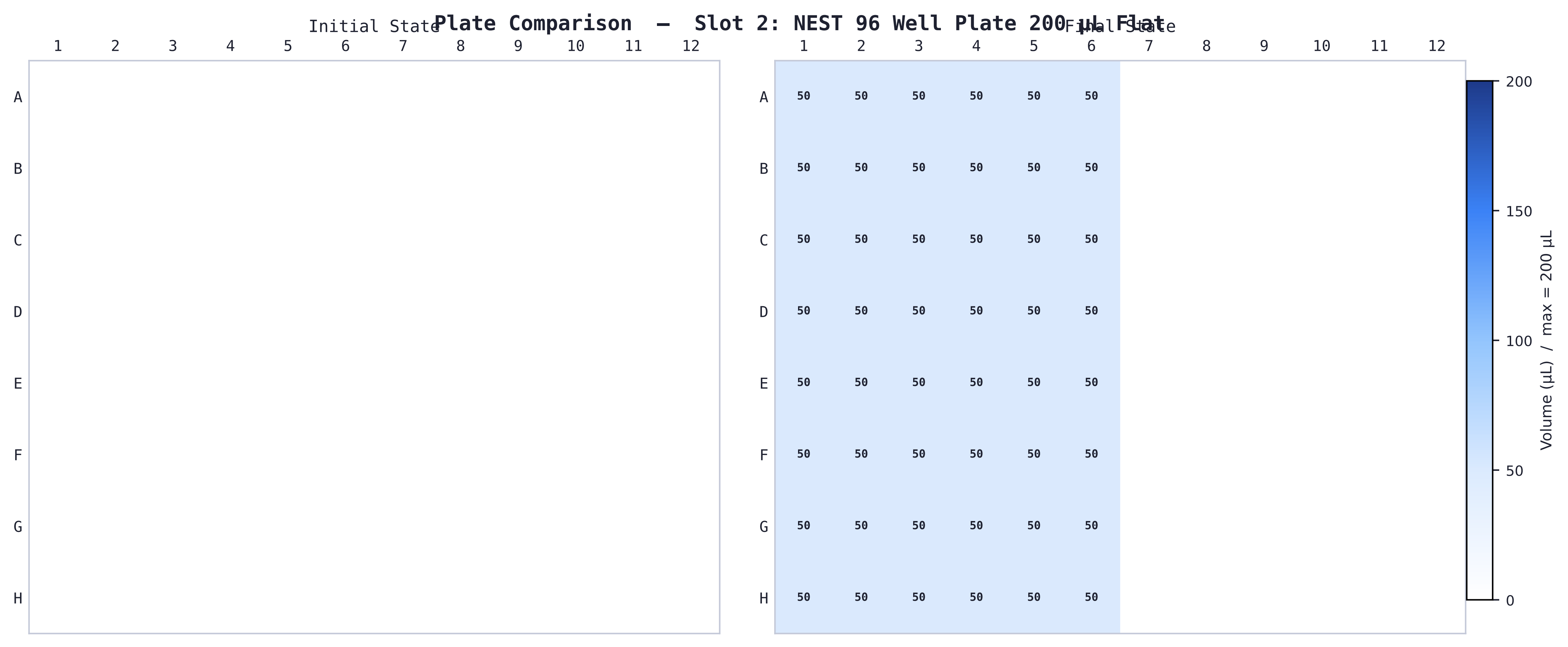


Plate Comparison — Slot 2: NEST 96 Well Plate 200 µL Flat
Initial State
Final State
1 2 3 4 5 6 7 8 9 10 11 12
A B C D E F G H
50
Volume (µL) / max = 200 µL
0 50 100 150 200

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Step	Action	Vol (µL)	Slot	Well	Labware
0	Pick Up Tip	–	1	A1	Opentrons OT-2 96 Tip Rack 300 µL
1	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
2	Dispense	50.0	2	A1	NEST 96 Well Plate 200 µL Flat
3	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
4	Pick Up Tip	–	1	B1	Opentrons OT-2 96 Tip Rack 300 µL
5	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
6	Dispense	50.0	2	B1	NEST 96 Well Plate 200 µL Flat
7	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
8	Pick Up Tip	–	1	C1	Opentrons OT-2 96 Tip Rack 300 µL
9	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
10	Dispense	50.0	2	C1	NEST 96 Well Plate 200 µL Flat
11	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
12	Pick Up Tip	–	1	D1	Opentrons OT-2 96 Tip Rack 300 µL
13	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
14	Dispense	50.0	2	D1	NEST 96 Well Plate 200 µL Flat
15	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
16	Pick Up Tip	–	1	E1	Opentrons OT-2 96 Tip Rack 300 µL
17	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
18	Dispense	50.0	2	E1	NEST 96 Well Plate 200 µL Flat
19	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
20	Pick Up Tip	–	1	F1	Opentrons OT-2 96 Tip Rack 300 µL
21	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
22	Dispense	50.0	2	F1	NEST 96 Well Plate 200 µL Flat
23	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
24	Pick Up Tip	–	1	G1	Opentrons OT-2 96 Tip Rack 300 µL
25	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
26	Dispense	50.0	2	G1	NEST 96 Well Plate 200 µL Flat
27	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
28	Pick Up Tip	–	1	H1	Opentrons OT-2 96 Tip Rack 300 µL
29	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
30	Dispense	50.0	2	H1	NEST 96 Well Plate 200 µL Flat
31	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
32	Pick Up Tip	–	1	A2	Opentrons OT-2 96 Tip Rack 300 µL
33	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
34	Dispense	50.0	2	A2	NEST 96 Well Plate 200 µL Flat

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Step	Action	Vol (µL)	Slot	Well	Labware
35	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
36	Pick Up Tip	–	1	B2	Opentrons OT-2 96 Tip Rack 300 µL
37	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
38	Dispense	50.0	2	B2	NEST 96 Well Plate 200 µL Flat
39	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
40	Pick Up Tip	–	1	C2	Opentrons OT-2 96 Tip Rack 300 µL
41	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
42	Dispense	50.0	2	C2	NEST 96 Well Plate 200 µL Flat
43	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
44	Pick Up Tip	–	1	D2	Opentrons OT-2 96 Tip Rack 300 µL
45	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
46	Dispense	50.0	2	D2	NEST 96 Well Plate 200 µL Flat
47	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
48	Pick Up Tip	–	1	E2	Opentrons OT-2 96 Tip Rack 300 µL
49	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
50	Dispense	50.0	2	E2	NEST 96 Well Plate 200 µL Flat
51	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
52	Pick Up Tip	–	1	F2	Opentrons OT-2 96 Tip Rack 300 µL
53	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
54	Dispense	50.0	2	F2	NEST 96 Well Plate 200 µL Flat
55	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
56	Pick Up Tip	–	1	G2	Opentrons OT-2 96 Tip Rack 300 µL
57	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
58	Dispense	50.0	2	G2	NEST 96 Well Plate 200 µL Flat
59	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
60	Pick Up Tip	–	1	H2	Opentrons OT-2 96 Tip Rack 300 µL
61	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
62	Dispense	50.0	2	H2	NEST 96 Well Plate 200 µL Flat
63	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
64	Pick Up Tip	–	1	A3	Opentrons OT-2 96 Tip Rack 300 µL
65	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
66	Dispense	50.0	2	A3	NEST 96 Well Plate 200 µL Flat
67	Drop Tip	–	12	–	A1 of Opentrons Fixed Trash
68	Pick Up Tip	–	1	B3	Opentrons OT-2 96 Tip Rack 300 µL
69	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL

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Step	Action	Vol (µL)	Slot	Well	Labware
70	Dispense	50.0	2	B3	NEST 96 Well Plate 200 µL Flat
71	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
72	Pick Up Tip	—	1	C3	Opentrons OT-2 96 Tip Rack 300 µL
73	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
74	Dispense	50.0	2	C3	NEST 96 Well Plate 200 µL Flat
75	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
76	Pick Up Tip	—	1	D3	Opentrons OT-2 96 Tip Rack 300 µL
77	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
78	Dispense	50.0	2	D3	NEST 96 Well Plate 200 µL Flat
79	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
80	Pick Up Tip	—	1	E3	Opentrons OT-2 96 Tip Rack 300 µL
81	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
82	Dispense	50.0	2	E3	NEST 96 Well Plate 200 µL Flat
83	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
84	Pick Up Tip	—	1	F3	Opentrons OT-2 96 Tip Rack 300 µL
85	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
86	Dispense	50.0	2	F3	NEST 96 Well Plate 200 µL Flat
87	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
88	Pick Up Tip	—	1	G3	Opentrons OT-2 96 Tip Rack 300 µL
89	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
90	Dispense	50.0	2	G3	NEST 96 Well Plate 200 µL Flat
91	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
92	Pick Up Tip	—	1	H3	Opentrons OT-2 96 Tip Rack 300 µL
93	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
94	Dispense	50.0	2	H3	NEST 96 Well Plate 200 µL Flat
95	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
96	Pick Up Tip	—	1	A4	Opentrons OT-2 96 Tip Rack 300 µL
97	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
98	Dispense	50.0	2	A4	NEST 96 Well Plate 200 µL Flat
99	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
100	Pick Up Tip	—	1	B4	Opentrons OT-2 96 Tip Rack 300 µL
101	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
102	Dispense	50.0	2	B4	NEST 96 Well Plate 200 µL Flat
103	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
104	Pick Up Tip	—	1	C4	Opentrons OT-2 96 Tip Rack 300 µL

Step Log (page 4 of 6)

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Step	Action	Vol (µL)	Slot	Well	Labware
105	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
106	Dispense	50.0	2	C4	NEST 96 Well Plate 200 µL Flat
107	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
108	Pick Up Tip	—	1	D4	Opentrons OT-2 96 Tip Rack 300 µL
109	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
110	Dispense	50.0	2	D4	NEST 96 Well Plate 200 µL Flat
111	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
112	Pick Up Tip	—	1	E4	Opentrons OT-2 96 Tip Rack 300 µL
113	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
114	Dispense	50.0	2	E4	NEST 96 Well Plate 200 µL Flat
115	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
116	Pick Up Tip	—	1	F4	Opentrons OT-2 96 Tip Rack 300 µL
117	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
118	Dispense	50.0	2	F4	NEST 96 Well Plate 200 µL Flat
119	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
120	Pick Up Tip	—	1	G4	Opentrons OT-2 96 Tip Rack 300 µL
121	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
122	Dispense	50.0	2	G4	NEST 96 Well Plate 200 µL Flat
123	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
124	Pick Up Tip	—	1	H4	Opentrons OT-2 96 Tip Rack 300 µL
125	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
126	Dispense	50.0	2	H4	NEST 96 Well Plate 200 µL Flat
127	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
128	Pick Up Tip	—	1	A5	Opentrons OT-2 96 Tip Rack 300 µL
129	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
130	Dispense	50.0	2	A5	NEST 96 Well Plate 200 µL Flat
131	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
132	Pick Up Tip	—	1	B5	Opentrons OT-2 96 Tip Rack 300 µL
133	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
134	Dispense	50.0	2	B5	NEST 96 Well Plate 200 µL Flat
135	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
136	Pick Up Tip	—	1	C5	Opentrons OT-2 96 Tip Rack 300 µL
137	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
138	Dispense	50.0	2	C5	NEST 96 Well Plate 200 µL Flat
139	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash

Step	Action	Vol (µL)	Slot	Well	Labware
140	Pick Up Tip	—	1	D5	Opentrons OT-2 96 Tip Rack 300 µL
141	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
142	Dispense	50.0	2	D5	NEST 96 Well Plate 200 µL Flat
143	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
144	Pick Up Tip	—	1	E5	Opentrons OT-2 96 Tip Rack 300 µL
145	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
146	Dispense	50.0	2	E5	NEST 96 Well Plate 200 µL Flat
147	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
148	Pick Up Tip	—	1	F5	Opentrons OT-2 96 Tip Rack 300 µL
149	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
150	Dispense	50.0	2	F5	NEST 96 Well Plate 200 µL Flat
151	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
152	Pick Up Tip	—	1	G5	Opentrons OT-2 96 Tip Rack 300 µL
153	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
154	Dispense	50.0	2	G5	NEST 96 Well Plate 200 µL Flat
155	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
156	Pick Up Tip	—	1	H5	Opentrons OT-2 96 Tip Rack 300 µL
157	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
158	Dispense	50.0	2	H5	NEST 96 Well Plate 200 µL Flat
159	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
160	Pick Up Tip	—	1	A6	Opentrons OT-2 96 Tip Rack 300 µL
161	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
162	Dispense	50.0	2	A6	NEST 96 Well Plate 200 µL Flat
163	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
164	Pick Up Tip	—	1	B6	Opentrons OT-2 96 Tip Rack 300 µL
165	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
166	Dispense	50.0	2	B6	NEST 96 Well Plate 200 µL Flat
167	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
168	Pick Up Tip	—	1	C6	Opentrons OT-2 96 Tip Rack 300 µL
169	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
170	Dispense	50.0	2	C6	NEST 96 Well Plate 200 µL Flat
171	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
172	Pick Up Tip	—	1	D6	Opentrons OT-2 96 Tip Rack 300 µL
173	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
174	Dispense	50.0	2	D6	NEST 96 Well Plate 200 µL Flat

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Step	Action	Vol (µL)	Slot	Well	Labware
175	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
176	Pick Up Tip	—	1	E6	Opentrons OT-2 96 Tip Rack 300 µL
177	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
178	Dispense	50.0	2	E6	NEST 96 Well Plate 200 µL Flat
179	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
180	Pick Up Tip	—	1	F6	Opentrons OT-2 96 Tip Rack 300 µL
181	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
182	Dispense	50.0	2	F6	NEST 96 Well Plate 200 µL Flat
183	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
184	Pick Up Tip	—	1	G6	Opentrons OT-2 96 Tip Rack 300 µL
185	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
186	Dispense	50.0	2	G6	NEST 96 Well Plate 200 µL Flat
187	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash
188	Pick Up Tip	—	1	H6	Opentrons OT-2 96 Tip Rack 300 µL
189	Aspirate	50.0	4	A1	NEST 1 Well Reservoir 195 mL
190	Dispense	50.0	2	H6	NEST 96 Well Plate 200 µL Flat
191	Drop Tip	—	12	—	A1 of Opentrons Fixed Trash